Rain, heat, and evolution

One of the most widely recognized systems of classification is the life zones system devised by American botanist Leslie Holdridge in 1947, and updated in 1967. His system is based on the assumption that two key factors, rain and heat, determine vegetation type in each region. He created a graphic depiction of 38 life zones in a pyramid. The three sides of the pyramid represent three axes: rain, temperature, and evapotranspiration (which depends on both rain and temperature). Using these axes, he could plot hexagons showing regions that also reflect humidity, latitude, and altitude.

American plant ecologist Robert Whittaker devised a much simpler graph, with average temperature on one axis and annual rainfall on the other. With these two variables plotted against each other, he was able to divide the graph into nine biomes—from tropical rain forest (the hottest and wettest) through to tundra (the coldest and driest).

Underpinning all these systems is the idea of convergent evolution, which argues that species develop similar traits as they adapt to similar environments. Insects, birds, bats, and pterosaurs all developed wings independently to occupy air space. Different biomes are therefore assumed to develop corresponding life forms in response to similar environmental conditions. However, in recent decades, it has been noted that species can evolve differently in the same biome and also that different stable biomes can develop in an identical climate. While central to understanding life, biomes remain a complex and elusive concept.